Carbon Capture Retrofit Tools

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Motivation

- Low TRL level technologies need to be evaluated and compared to DOE goals • Full evaluations of power plants with new technologies integrated take time and resources
- Develop high level capability to assess CO_2 capture costs to existing fleet of CO_2 sources including PC, NGCC and Industrial Sources

CO_2 Capture Methodology Spreadsheet

- Input: performance and cost estimates from developer models
- Spreadsheet estimates performance and cost of base plant and provides overall plant metrics
- Performance and cost calculations based upon model developed from Bituminous Baseline Study, Rev 3 Case B12B
- Detailed description of model contained in methodology document

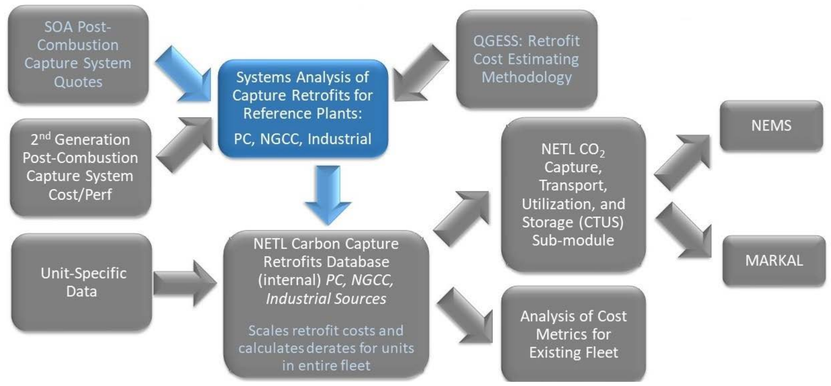
	A	B	C	D	E	F
1	POST-COMBUSTION CO_2 CAPTURE TECHNOLOGY ASSESSMENT METHODOLOGY					
54						
55	BASE PLANT CO_2 CAPTURE PROCESS PERFORMANCE & COST INPUTS		Input Cells	Calculation Cells	PRESS F9 TO RECALCULATE	
56	**Developing CO_2 Capture Technology Identifier**	Base Case Example	Adsorbent	Membrane		
57		BBR B12B Cansolv	Example	Example		
58	**CO_2 Capture Process Performance Inputs (Inputs generated using prescribed design basis)**					
59	No. 1 Extracted steam heating duty: Q_{e1} (MMBtu/hr)	1,129	1,000	0		
60	No. 1 Extracted steam heating temperature (F)	293.5	284	230		
61	No. 2 Extracted steam heating duty: Q_{e2} (MMBtu/hr)	0	0	0		
62	No. 2 Extracted steam heating temperature (F)	230	230	230		
63	CO_2 Capture Process cooling duty: Q_c (MMBtu/hr)	1,868	200	350		
64	CO_2 Separation System pressure drop: DP_{SEP} (psi)	0	0.9	2		
65	FD-fan pressure boost above Base value of 0.6 psia: DP_{FDfan} (psi)	0	0	1		
66	CO_2 Capture Process CO_2 removal efficiency (%)	90	91	90		
67	CO_2 Separation System power: A_{SEP} (kW)	16,000	10,900	30,000		
68	CO_2 CPU power: A_{CPU} (kW)	35,690	49,000	51,000		
69	**CO_2 Capture Process Cost Inputs (Inputs generated using prescribed design basis)**					
70	CO_2 Separation System cost ($1000: Cost Base June 2011, TPC basis)	533,757	400,000	350,000		
71	CO_2 CPU cost ($1000; Cost Base June 2011, TPC basis)	98,381	100,000	140,000		
72	Key CO_2 Capture Process variable cost items	Makeup solvent	Makeup adsorbent	Membrane surface		
73	CO_2 Capture Process variable cost: C_{VarCO2} ($1000/yr)	0	5,000	8,000		

CO_2 Purification and Compression Spreadsheet

- Input: composition and conditions of CO_2 product from the capture system
- Spreadsheet estimates performance and cost of CO_2 compression and (if required) CO_2 purification system
- Performance and cost calculations based upon Aspen model of CO_2 purification and compression system for performance and Aspen Economic Analyzer for costs
- Results can be used as inputs to CO_2 Methodology spreadsheet
- Detailed description of model contained in methodology document

	A	B	C	D	E
1	RAW CO_2 GAS FINAL PROCESSING SYSTEM PERFORMANCE AND COST SPREADSHEET				
2	Application	PC Post-Combustion (coal rate fixed at BBR Case B12B)			
3					
4		STUDY PROCESS	Correlation Limits	BASELINE PROCESS	
5					
6	Raw CO_2 Gas Composition (mole fraction)				
7	CO_2	0.6	OK	0.9824	
8	H_2O	0.3	OK	0.0176	Developer Input Data for Raw CO_2 Gas
9	O_2	0	OK	0	Calculated Results
10	N_2	0.1	OK	0	PRESS F9 TO RECALCULATE
11	Ar	0		0	
12	Total	1		1	
13	Raw CO_2 Gas Inlet Pressure (psia)	15	OK	28.7	
14	Raw CO_2 Gas Inlet Temperature (°F)	85	OK	86	
15	Separation System CO_2 Efficiency (%)	90.0		91.5	
16					
17	Raw CO_2 Gas Flow (lbmol/hr)				
18	CO_2	24,060		24,461	
19	H_2O	12,030		438	
20	O_2	0		0	
21	N_2	4,010		0	
22	Ar	0		0	
23	Total	40,100		24,899	
24					
25	Raw CO_2 Gas Processing Needed	CPU		Compression	
26	Power Consumed (kW)	57,336		33,997	
27	Cooling-Water Load (MMBtu/hr)	537		230	
28	CO_2 Loss Rate (% of inlet CO_2)	6.5		0.0	
29	Separation System CO_2 Efficiency Needed (%)	96.3		90.0	
30	Total Plant Cost (Million June 2011$)	193		98	

Carbon Capture Retrofit Database (CCRD)



User Inputs and Constants Used in Calculations

Input Desired Scenarios to be Analyzed Here

Parameter	Units	Value SCENARIO #1	Value SCENARIO #2	Value SCENARIO #3	Comments
CO_2 Capture Rate	Choose option	Default	Default	Default	
	%	90%	90%	90%	
CO_2 Capture Technology	Choose Option	Amine Based	Amine Based	Amine Based	
Pre-Retrofit Capacity Factor	Choose Option	User Input	User Input	User Input	
	%	85%	75%	65%	Unit Actual uses plant level reported CFs. Average value is the average reported capacity factor from 2013-2015.
Post-Retrofit Capacity Factor	Choose Option	Delta	Delta	Delta	
	% or Reduction Delta	0%	10%	10%	Use Absolute or delta from pre-retrofit CF. If the delta entered is lower than the unit's actual CF, the plant will be excluded from calculations.
Retrofit Unit Capacity Applicability Limit	Choose Option	Default	Default	Default	
	MW	25	25	25	All units with a nameplate capacity below the limit are excluded from all calculations.
Retrofit Cost Factor	Choose Option	Default	Default	Default	
		1.10	1.10	1.10	Multiplier applied to Total Plant Capital Cost.
Capital Charge Factor	Choose Option	High Risk	High Risk	High Risk	
	Choose Option	30-year	30-year	30-year	Default values are for a 3-year construction period with either a 10-, 20-, or 30-year economic life, investor owned utility.
		0.111	0.111	0.111	
Advanced Options					
CO_2 Emissions Rate	Choose option	Unit Actual	Unit Actual	Unit Actual	
	lb/MMBtu	N/A	N/A	N/A	
Maximum CO_2 Capture Rate Per Train	Choose option	Default	Default	Default	
	TPD	15,772	15,772	15,772	If the CO_2 production rate exceeds the maximum, multiple trains will be used.
Plant Capacity Metric		Nameplate	Nameplate	Nameplate	
Cost Year Basis	Choose Option	Default	Default	Default	
	Year	2011	2011	2011	Scaled based on Chemical Engineering Cost Index (CEPCI). Only 2011 or 2017 can be selected
Cooling preference?	Choose Option	Existing	Existing	Existing	Select cooling system (dry or wet) based on either type of existing cooling system or location's (State or NERC Region) water availability.
Projected Sales Price of Electricity	Choose Option	Default	State	State	Cost for lost revenue/price to purchase make-up power. State uses annual average retail price of electricity to ultimate customers, all sectors from EIA. Only used in breakeven CO_2 metric calculations.
	$/MWh	60	N/A	N/A	
Include SCR with retrofit?	Choose Option	Yes	Yes	Yes	
Include FGD with retrofit?	Choose Option	Yes	Yes	Yes	
Additional Heat Rate Penalty	Choose Option	None	None	None	
	Btu/kWh	N/A	N/A	N/A	This value is a user specified penalty in addition to the calculated values for CO2 capture and selected optional systems (SCR and FGD).
CO_2 Transport and Storage Costs	Choose Option	Default	Default	Default	
	$/tonne captured	11.00	11.00	11.00	COE and Breakeven CO_2 Emissions Penalty include T&S. Breakeven CO_2 Sales Price never includes T&S.

CCRD Overview

- Evaluate scenarios where there is a price on CO_2 emissions
- Consider economic feasibility of power plants or industrial sources that sell CO_2 for Enhanced Oil Recovery (EOR)
- Analyze benefits of CO_2 capture R&D on existing fleet of power plants and industrial sources

- Separate tools available for Pulverized Coal, Natural Gas Combined Cycle and Industrial Sources
- PC and NGCC tools pre-populated with NETL Baseline Study cases for reference
  - User may provide plant specific data from other sources
- Facility data for industrial sources based on EPA's Greenhouse Gas Reporting Program (http://www.epa.gov/ghgreporting/ghgrp-reported-data) and FLIGHT data (http://ghgdata.epa.gov/ghgp/main.do#)

- CCRD Tools and Reference Documents available at the following site when searching for **Collection Name – Carbon Capture Retrofit Database**
  - http://www.netl.doe.gov/energy-analysis/search

CCRD Example Results

